

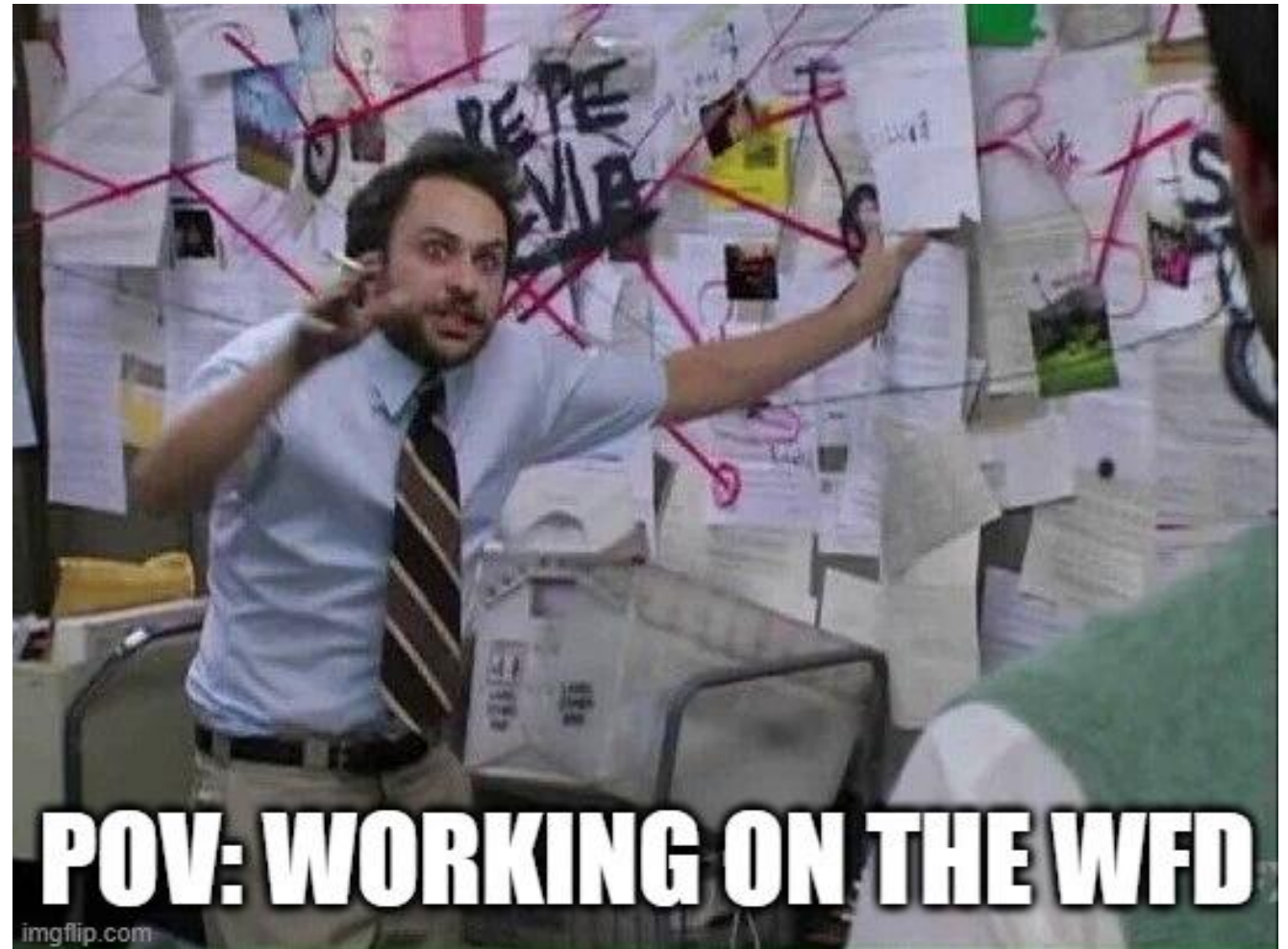
Tips & Tops

PM/SE Tasks Project Plan

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Project Plan PMSE Deliverables

1. Work Breakdown Structure (WBS)
2. Workflow Diagram (WFD)
3. Gantt Chart
4. Organogram (Team Organisation Structure)
5. Organisational Risk Analysis

Recommended Software/Tools

1. Work Breakdown Structure (WBS)

- Diagrams.net (Recommended)
- Miro

2. Workflow Diagram (WFD)

- Diagrams.net (Recommended)
- Miro

3. Gantt Chart

- Online Software: InstaGantt, Quire, Notion → *before choosing one, check if and how exporting a diagram is possible!*
- Coding/Programming: Matplotlib, PlantUML

4. Organogram (Team Organisation Structure)

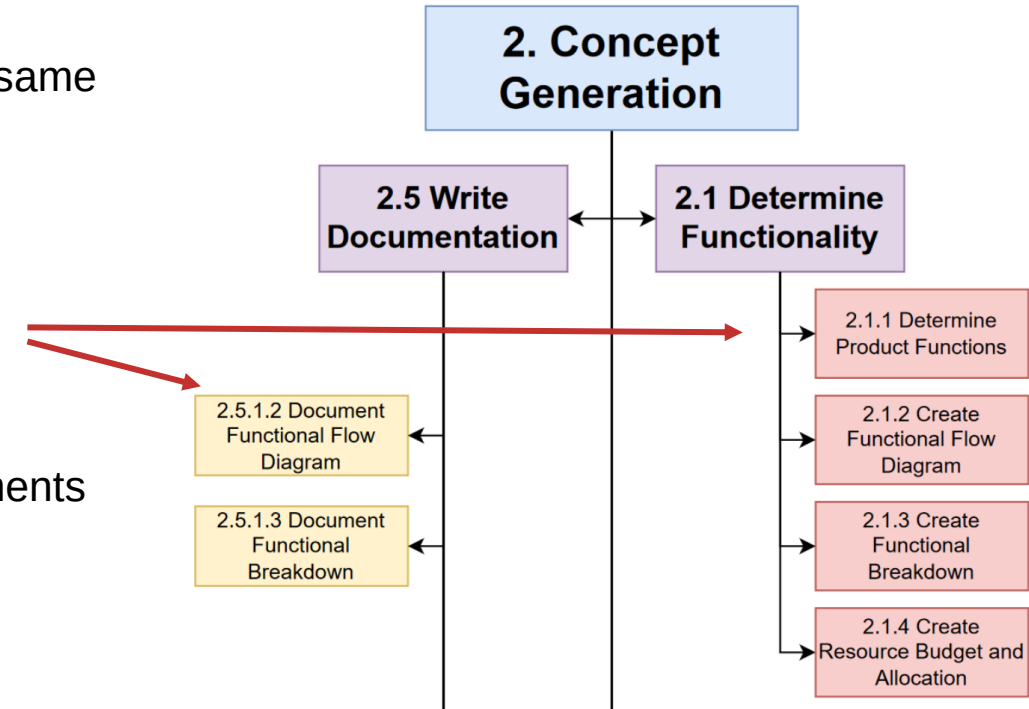
- Diagrams.net (Recommended)
- Miro

5. Organisational Risk Analysis

- n/a

Workflow Diagram (WFD)

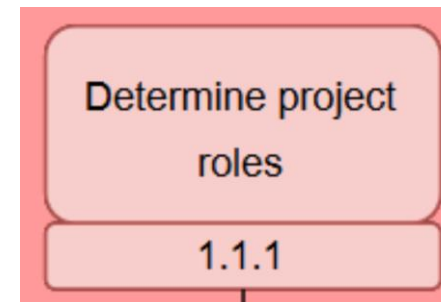
- Identifiers (IDs) should represent the hierarchical relation of tasks with respect to each other.
 - They should be consistent across diagrams when presenting the same tasks (WBS & WFD).
 - IDs and numbering should also be consistent across ALL DSE documents and deliverables!
 - Chosen format for the IDs should be clearly defined in text.
 - Should have the same format for tasks at the same level of detail.
- A legend should be present and should fully describe the visual elements used.
 - “Special” blocks and symbols should be clearly defined.
 - Logical operators should be clearly defined (such as AND/OR/CONDITIONS/etc..).
 - Chosen block format for tasks should be clearly defined



[Sample from a WBS (not a WFD) for better clarity of hierarchy and task relations]

Workflow Diagram (WFD)

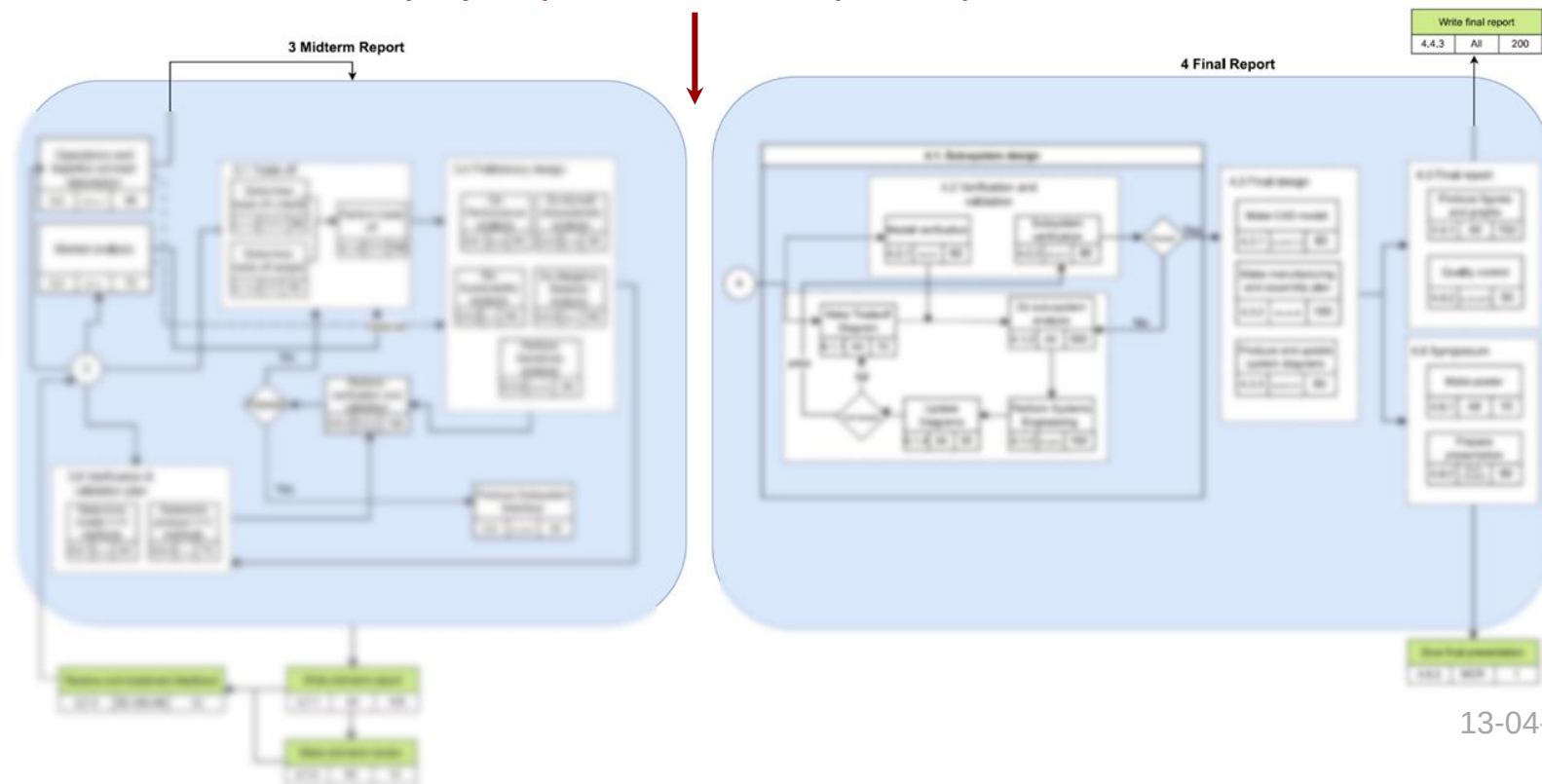
- General common mistakes:
 - Convoluted formatting, flow from one task to the next is not clear
 - Insufficient graphical separation between hierarchical levels of tasks
 - Tasks are only sequential, it's important to show the parallel nature of tasks and the dependencies between them
 - Lack of iterative relations, many tasks and project elements will be iterative, show that.
 - Lacking organisational tasks, do not forget tasks such as quality control, processing feedback, preparing final report version, submitting final report version, etc.
 - Break down deliverables further, do not present “*make gantt chart*” as a single task, more detail must be added, consider what tasks should be completed to *make the gantt chart*.
- Include all deliverables: Personal Appendix, Red Booklet, Poster, Symposium, Jury Summary, etc.
- Add attributes to task blocks where useful:
 - ID [required],
 - Responsible person [required]
 - Effort (hours) [required]
 - Throughput time (hours) [recommended]



- Identifier
- Responsible person
- Effort (hours)

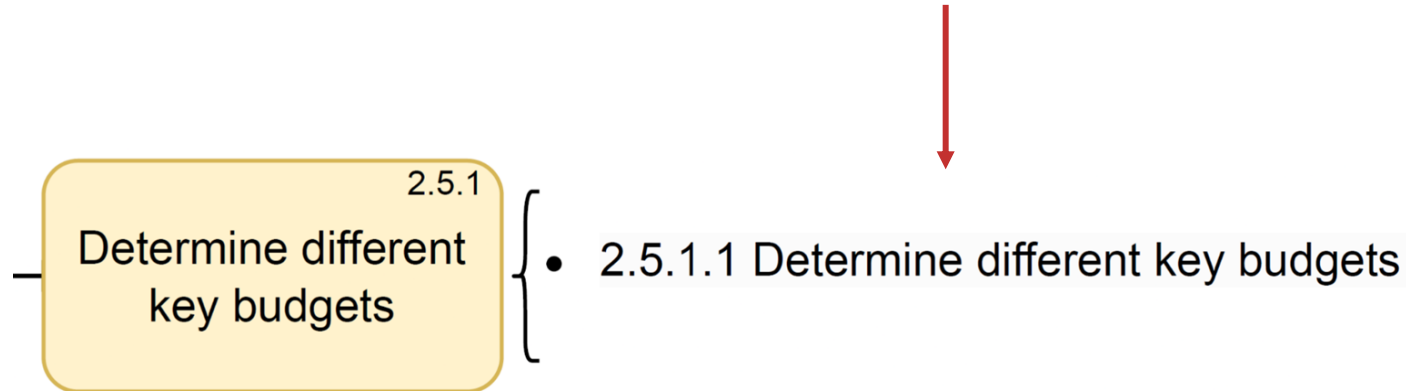
Workflow Diagram (WFD)

- DSE deliverables are **NOT** project phases!
 - For example, midterm report is the final deliverable of the conceptual design and trade-off phase.
- Diagram should present an unbroken logical flow of sequential and parallel tasks.
 - ↳ Missing connections between separate project phases. (Note that there could be multiple connections between two project phases based upon dependencies.)



Work Breakdown Structure (WBS)

- 3 levels of detail are required for all tasks up to and including the *Conceptual Design and Trade-Off Phase* (with *Midterm Report* as final deliverable), not required for *Preliminary Design Phase* → Level of detail should be sufficient such that the lowest level tasks can be completed within 4-8 hours, if not, further subdivide defined tasks.
- All 2nd-level tasks require at least two 3rd-level task beneath them. Having only one would be equivalent to rephrasing the 2nd-level task.



Work Breakdown Structure (WBS)

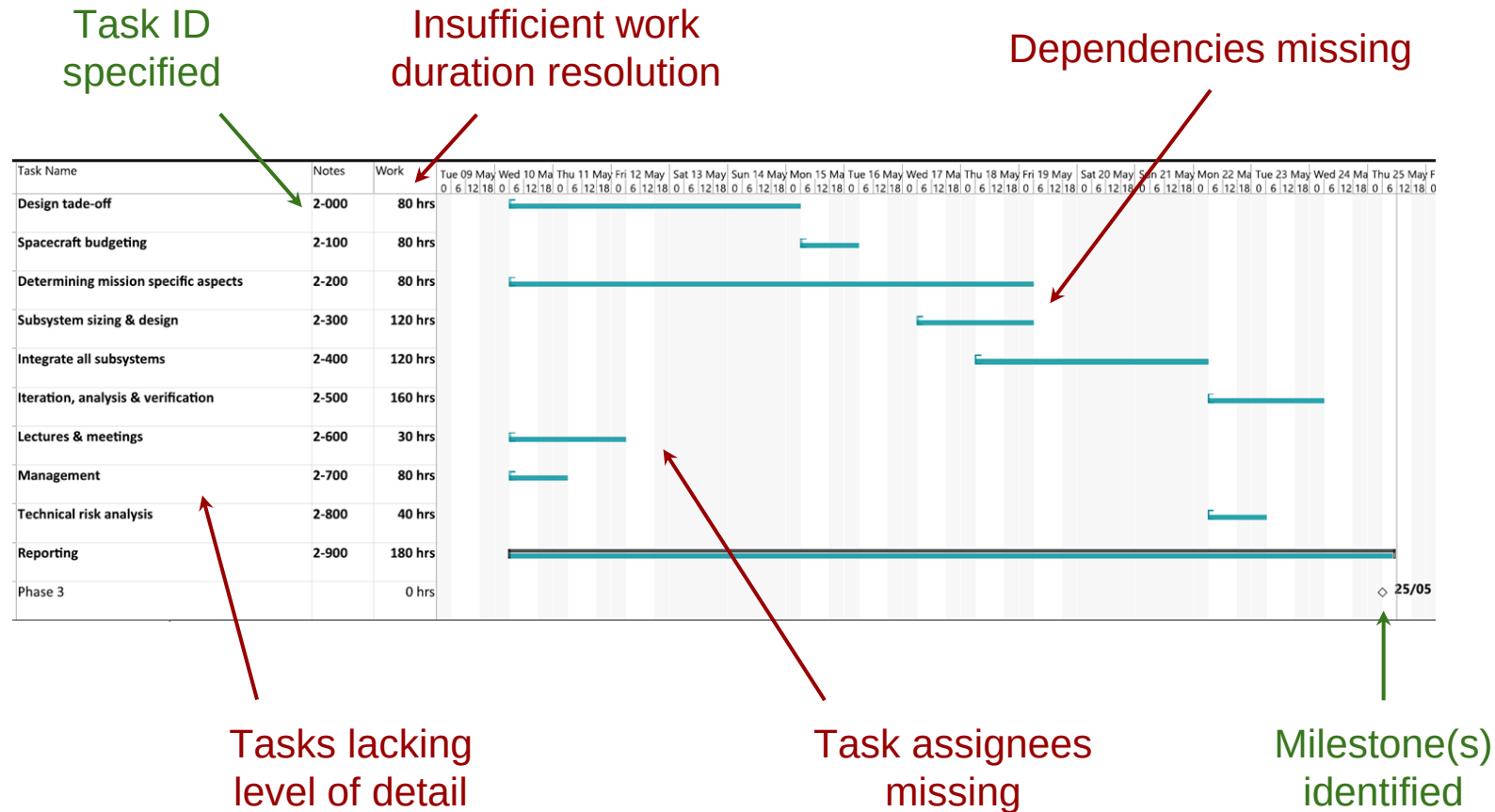
- Be consistent across diagrams: tasks in the WBS should be identical to the ones in the WFD.
- Include all deliverables: Red Booklet, Personal Appendix, Poster, Symposium.
- Each task should include an identifier, responsible person, and effort in hours (like in the WFD).
- Legend should fully define the diagram visual elements
- Relevant parameters definitions:
 - **Throughput time** → time it will take for the task to be completed
 - **Effort/Work hours** → total number of hours if one person were to do the whole task
 - **Responsible person** → single person holding responsibility for the task
 - **Assigned person/people** → all people who will work on the task

Gantt Chart

- Dependencies between tasks should always be present.
 - This helps you determine the most efficient scheduling of your tasks - what is sequential and what can be done in parallel.
 - It is important to identify which tasks flow into one another, such that if one task is delayed, it can be determined which subsequent tasks are affected and how to reschedule them.
 - **It is not possible to schedule a task to start before the completion of a task that it is dependent upon.**
- Aim for half-day resolution in the Gantt chart.
- Assignees to tasks should be clear. (all people assigned)
- At least the same level of detail of the WBS. If not half day resolution, break task down further.

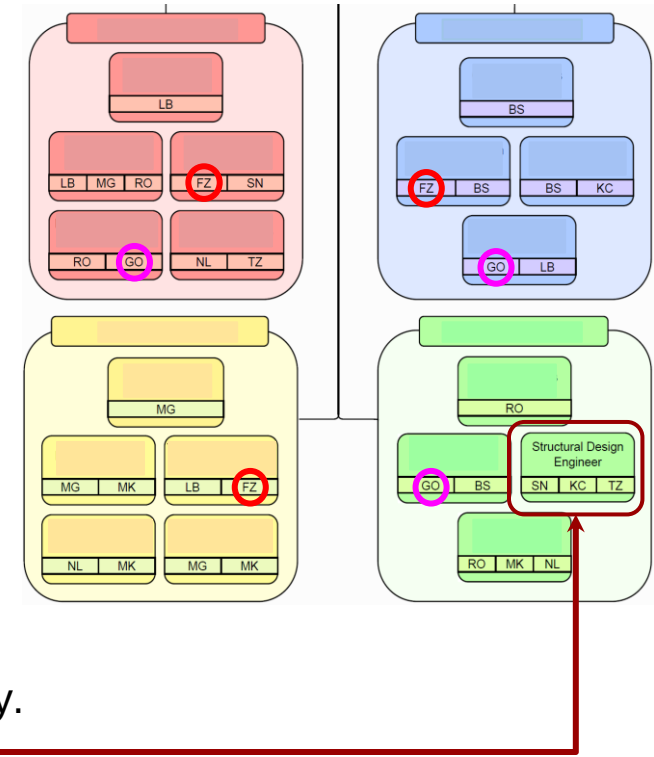


Gantt Chart



Organogram

- Every member should have one *organisational* and one *technical* role.
 - The responsibilities of the roles should be described (e.g. in a Table).
 - Try to keep people assigned to a specific department (maximum 2), otherwise they will need to keep up to date on the development of multiple subsystems which adds additional overhead.
 - Important to identify backup roles in case a member is absent/falls ill.
 - People should not be assigned to multiple roles and to multiple departments. →
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- Make the roles explicit: only one person should be assigned to each responsibility.
 - Multiple people should not be assigned to the same position. —————
 - Multiple responsible people for the same position does not make sense → clear distinction between responsible person and everyone who will work on something, the organogram should show responsibilities



Organogram

- Identify the interface between the organisational and technical departments through the chief/systems engineer.
- Inclusion of portraits for the roles is appreciated!
- Clear legend to interpret visual diagram should be present.
- Avoid having a chief engineer **AND** a system engineer (mandatory). Too much overlap for the project scope.
- Avoid combining the project manager and the chair of meetings
- It is advised to discuss roles and potentially re-assign them after the midterm deliverable. This allows group members to have experience with the different responsibilities.

Organisational Risk Analysis

- Organisational SWOT analysis is the first step of the risk analysis, ensuring all possible risks are identified and considered.
- Clear definition of what likelihood and consequence mean in the risk map.
- Explain what the levels of likelihood and consequence mean.
- Risks presented should have a mitigation strategy and a **contingency plan** in case it still occurs
- Clear distinction between preventive measures to mitigate risks, and contingency plans if risk materialises
- Two risk maps should be present: pre-mitigation and post-mitigation.
- Risks should possess an ID and be assigned a specific member that is responsible for monitoring each risk (the risk manager cannot be assigned to all risks).

		Likelihood and Consequence levels missing		Mitigation strategy present	Risk post-mitigation missing	Specific member assigned (name preferred)
ID missing		Risk	Schedule impact	Mitigation action	Responsible member	
1	Unavailability of tutor	0.2	Interruption of work	Work ahead on other tasks.	Secretary	
2	Holdup time for parallel tasks	0.2	Interruption of work	Use preliminary (estimated) values, iterate with final values. Else continue with other tasks.	Chief Engineer	



Good luck!

Alessandro and Austin